Finite-Size Majorana Splitting in the Kitaev Chain: Why do the "spikes" in the finite-size spectrum worsen with larger L?

Seminar notes

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1 Setup

We study the Kitaev chain at $t = \Delta = 1$ with open boundary conditions (OBC). The BdG Hamiltonian is the real-symmetric matrix

$$M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \qquad h_{ij} = -\mu\delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \qquad d_{ij} = \Delta(\delta_{i,j+1} - \delta_{i+1,j}). \tag{1}$$

Particle-hole symmetry (PHS) guarantees eigenvalues come in exact $\pm E$ pairs. Plot 4 tracks the *six smallest positive eigenvalues* as μ varies.

2 The ideal edge-mode energy

2.1 Zero-energy condition in the complex BZ

To find the edge-mode energy, we look for a zero-energy solution that decays into the bulk. Set $E = 0$ and try an Ansatz $\psi_j \sim \lambda^j$. Substituting into the bulk BdG equations gives the characteristic equation

$$\left(-\mu - t(\lambda + \lambda^{-1})\right)^2 + \Delta^2(\lambda - \lambda^{-1})^2 = 0. \tag{2}$$

For $t = \Delta$ this factors to

$$\left(-\mu - t(\lambda + \lambda^{-1}) + \Delta(\lambda - \lambda^{-1})\right)\left(-\mu - t(\lambda + \lambda^{-1}) - \Delta(\lambda - \lambda^{-1})\right) = 0. \tag{3}$$

The two branches give the solutions

$$\lambda_A = \frac{\mu}{2t}, \qquad \lambda_B = \frac{2t}{\mu}. \tag{4}$$

For the left edge mode we need $|\lambda| < 1$, so we take $\lambda = \mu/(2t)$ (valid for $|\mu| < 2t$, i.e. inside the topological phase). The wavefunction decays *purely exponentially*:

$$\psi_j^L \sim \left(\frac{|\mu|}{2t}\right)^{j-1}, \qquad \xi = \frac{1}{|\ln|\mu/2t||} \quad \text{(coherence length)}. \tag{5}$$

Crucially, for $t = \Delta$ the solution has **no oscillatory factor** $e^{ik_0 j}$. This is special to the symmetric point; for $t \neq \Delta$ the roots can be complex, introducing oscillations.

2.2 Finite-size hybridization splitting

A chain of length L has both a left edge mode $\psi_j^L \sim \lambda^{j-1}$ and a right edge mode $\psi_j^R \sim \lambda^{L-j}$. Their overlap determines the splitting of the zero-energy pair:

$$\delta E \approx 2t \, \frac{\langle \psi^L | V | \psi^R \rangle}{\|\psi^L\| \, \|\psi^R\|} \sim 2t \, \lambda^L = 2t \left(\frac{|\mu|}{2t} \right)^L. \tag{6}$$

This is the energy of the near-zero mode. For $t = \Delta = 1$:

$$\boxed{E_0(\mu, L) \approx 2 \left(\frac{|\mu|}{2} \right)^L.} \tag{7}$$

Predictions from (7):

- At the sweet spot $\mu = 0$: $E_0 = 0$ *exactly*, for any L.
- At the phase boundary $|\mu| = 2t$: $E_0 \to 2$ (the bulk gap), for any L.
- In the interior: E_0 decreases *exponentially* with L. A larger chain has a better-protected zero mode.

So (7) predicts that larger L should give *smaller* spikes, not larger ones.

3 Why the spikes increase and widen with L: a numerical artifact

3.1 The floating-point threshold problem

The observed behaviour (spikes worsen as L increases) is *not* a physical effect — it is a numerical artefact arising from how we define "positive spectrum".

In the simulation, positive eigenvalues are selected with the filter `evals[evals >= 0]`. For large L, the ideal splitting $E_0 = 2(|\mu|/2)^L$ falls below machine precision ($\approx 10^{-15}$) over a wide range of μ:

$$E_0 < \varepsilon_{\text{mach}} \iff \left(\frac{|\mu|}{2} \right)^L < \frac{\varepsilon_{\text{mach}}}{2} \iff |\mu| < 2 \, \varepsilon_{\text{mach}}^{1/L}. \tag{8}$$

For $\varepsilon_{\text{mach}} = 5 \times 10^{-16}$:

$$L = 30: \quad |\mu|_{\text{thresh}} \approx 0.62, \tag{9}$$

$$L = 100: \quad |\mu|_{\text{thresh}} \approx 1.41. \tag{10}$$

For $|\mu| < |\mu|_{\text{thresh}}$, the near-zero eigenvalue is indistinguishable from zero at double precision. Due to rounding, it can be stored as a tiny *negative* number ($\sim -10^{-16}$). The filter `evals >= 0` then *excludes* this eigenvalue, and the *lowest bulk mode* (at finite energy $\sim E_{\text{gap}}$) takes its place as the "lowest positive eigenvalue". This appears as a spike.

3.2 Width and height of the artefact

- **Width**: the spike region is $|\mu| < 2\varepsilon_{\text{mach}}^{1/L}$, which *grows* with L (for $L = 100$ it covers most of the topological phase, explaining the "widened" spikes).
- **Height**: the spike jumps to the lowest bulk energy, which is $\sim E_{\text{gap}}(\mu)/(L\text{-dependent level spacing})$. With larger L the bulk level spacing is smaller, so more bulk levels fall inside the window and the spikes look denser.

4 The fix: use quasiparticle energies $|E|$

The physical quasiparticle spectrum consists of the *absolute values* of the BdG eigenvalues. The near-zero mode has $|E| \approx 0$ regardless of the sign of the floating-point result:

$$\text{correct}: \quad \{E_n\}_{n=1}^{L} = \text{sort}\big(|e_i|\big)_{i=1}^{2L}. \tag{11}$$

In code, replace

```
return evals[evals >= 0]          # WRONG: sign-sensitive
```

with

```
return np.sort(np.abs(evals))[:L]  # correct: quasiparticle energies
```

This is the standard way to extract the quasiparticle spectrum from a BdG diagonalisation and is immune to the sign ambiguity at near-zero eigenvalues.

5 Summary

Observation	Physical or artefact?	Cause
Spikes near $|\mu| \to 2t$	**Physical**	$\xi \to \infty$, hybridisation is strong
Spikes in interior for small L	Mostly artefact	Float precision, level crossings
Spikes widen/worsen for large L	**Artefact**	Threshold $|\mu|_{\text{thresh}} \propto \varepsilon^{1/L}$ grows